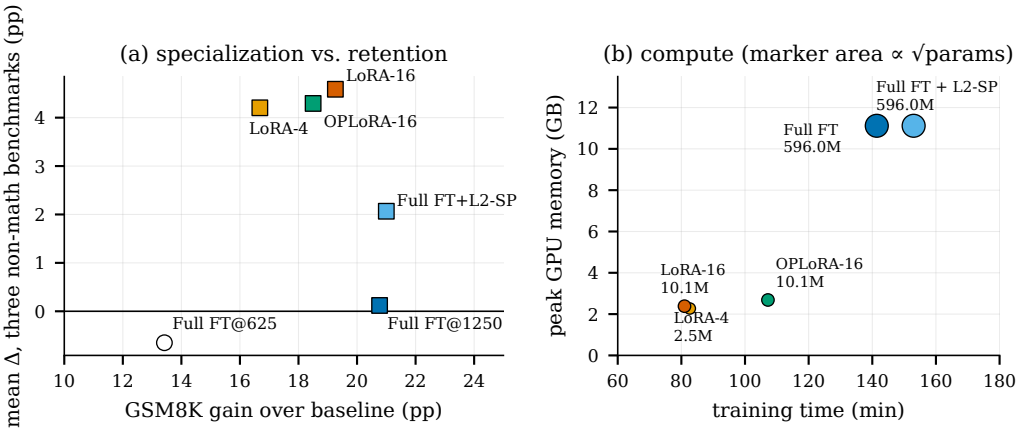
(a) specialization vs. retention
LoRA-16
LoRA-4
OPLoRA-16
Full FT+L2-SP
Full FT@625
Full FT@1250
mean Δ, three non-math benchmarks (pp)
GSM8K gain over baseline (pp)
(b) compute (marker area ∝ √params)
Full FT + L2-SP
596.0M
Full FT
596.0M
LoRA-16
10.1M
OPLoRA-16
10.1M
LoRA-4
2.5M
peak GPU memory (GB)
training time (min)